

Design and Verification of a Two-Chip-Cascaded 4D mm-Wave Imaging Radar for Indoor SLAM

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Abstract — Since the concept of simultaneous localization and mapping (SLAM) was proposed and investigated in 1980s, it has been attracting continuous attention in robotics regime. The sensor types applied for perception in SLAM are usually stereo or RGB cameras, also lidars based on point clouds output with high density and accuracy. Driven by the autonomous driving in the automobile industry emphasizing both performance and cost, the 4D millimeter-wave imaging radar becomes a new alternative for the SLAM perception. This work focuses on a SLAM oriented solution merely using the implemented mm-wave 4D imaging radar for an indoor demonstration. The two-chip cascaded 4D radar offers totally 6 transmitting and 8 receiving physic channels in RF front-end. By applying sparse antenna array and multi-input multi-output (MIMO) technique, 48 virtual channels can be emulated, which achieve angle separation each of 1.5° on the azimuth and 3° on the elevation plane. An indoor 3D SLAM demonstration has been also done based on real-time point clouds output of the implemented 4D imaging radar.

Keywords — SLAM, 4D, mm-wave, imaging radar, sparse antenna array, MIMO.

I. INTRODUCTION

The concept of simultaneous localization and mapping (SLAM) was originally proposed during the IEEE Robotics and Automation Conference in 1986, San Francisco, California [1]. Since then, SLAM has become one of the most challenging but fascinating research topics in robotics regime. SLAM solution endues a robot the capability to plan its own path in an autonomous manner.

Camera -based SLAM have paved the way with great success as virtual method dominates in robotics perception systems [2]- [7]. Despite high cost, lidar-based SLAM [8]- [11] also becomes as one promising approach due to recently emerging lidar application in autonomous driving functions. As the conventional advanced driver assistant system (ADAS) millimeter-wave (mm-wave) radar has begun its evolution towards 4D point cloud imaging phase [12]- [14], mm-wave radar-based SLAM solutions have been correspondingly brought under spotlight [15]-[17] to enhance system performance under bad weather condition (rainy, snowy, foggy...etc.) in autonomous driving functions. This work presents the design, implementation, and verification of a two-chip cascaded 4D mm-wave imaging radar, which is suitable for an indoor SLAM application. By applying sparse antenna array and multi-input multi-output (MIMO) technique, 48

virtual channels can be emulated based on physically existing 6 transmitting and 8 receiving channels, which achieve angle separation each of 1.5° on azimuth and 3° on elevation plane. With maximum detection range of 51 m, the implemented 4D radar can demonstrate reasonable imaging effect for SLAM application by generating 10240 points/s in various indoor scenarios. The demonstration showed that the implemented 4D mm-wave imaging radar could be well applied for pure radar-based indoor 3D SLAM applications.

II. SYSTEM DESIGN

A. System Block Diagram

Figure 1 shows the system block diagram of the implemented 4D radar sensor. It consists mainly of two TEF8232 radar transceiver front-end chips and S32R294 microprocessor of power PC architecture. Since two radar transceiver chips each with 3 transmitting and 4 receiving channels are cascaded in form of “Master” and “Slave” by sharing the common local oscillator (LO) and clock (CLK) signal, the overall system owns then synchronized 6 transmitting and 8 receiving channels.

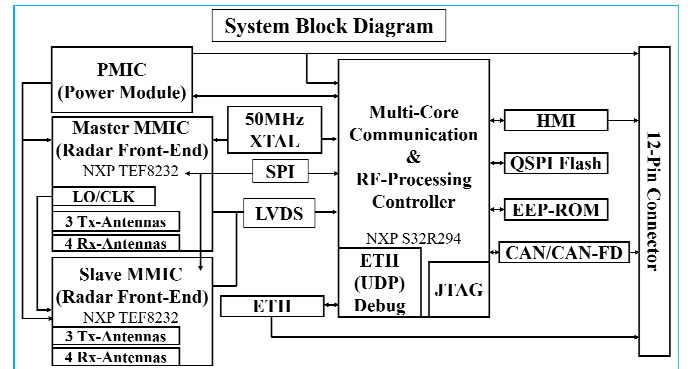


Fig. 1. System block diagram of the implemented two-chip-cascaded 4D mm-wave imaging radar sensor mainly based on two cascaded radar front-end chips (TEF8232) and one processing unit (S32R294).

B. Sparse Antenna Array

Figure 2 presents the antenna array design of the implemented 4D imaging radar sensor. With MIMO and sparse array technique, 6 transmitting antennas (Tx) and 8 receiving antennas (Rx) are so arranged on the printed circuits board

(PCB) in Fig. 2 (left), that the equivalent virtual antenna array can achieve the maximum aperture each of 25.5λ on azimuth and 19.25λ on elevation plane with λ of the wavelength in 77/79 GHz frequency band.

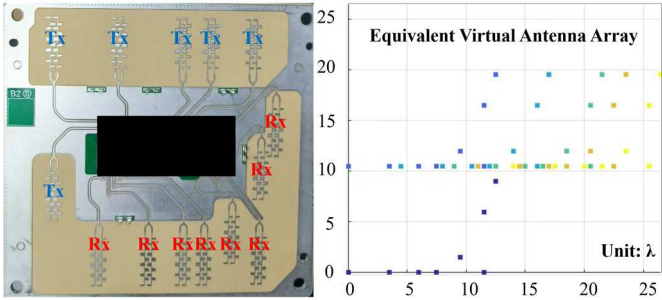


Fig. 2. Sparse array antenna distribution on implemented PCB with 6 transmitting and 8 receiving channels (left); equivalent virtual antenna array with 48 channels distributed on azimuth and elevation plane (right).

This implementation distributes all 48 virtual channels based on the following two principles: 1. Aperture on both azimuth and elevation plane should be larger than 15λ but less than 27λ (limited by the total radar size); 2. Angular separation of the implemented 4D imaging radar sensor on both azimuth and elevation plane should achieve 1.5° and 3° .

C. Mechanical Design Concept

The explosion view of the implemented 4D imaging radar sensor is shown in Fig. 3 (left) with the detailed dimension information in Fig. 3 (right). Millimeter-wave components such as cascaded radar chips, antenna array, power distribution networks, crystal are located on high frequency (HF) board. Processor chip, power management module, and all peripherals (flash, EEP-ROM, Ethernet, CAN, HMI...etc.) are populated on low frequency (LF) board.

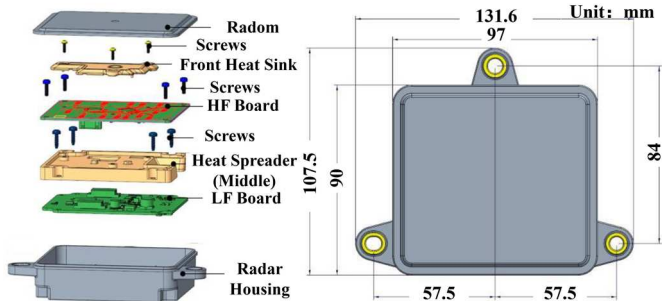


Fig. 3. Mechanical concept of the implemented 4D mm-wave imaging radar sensor (explosion view: left; dimensions: right).

Table 1. Angular resolution measurement on the azimuth and the elevation planes for the implemented 4D mm-wave imaging radar.

Set Value			4D Radar Under Test (RUT)	
Angle/Plane	Target RCS	Range	Distance of two corner reflectors	Angular separation
0°/Az.	10 dBsm	10 m	26.88 cm	1.44°
0°/El.	10 dBsm	10 m	50.28 cm	2.88°

Thermal management considered design is implemented via the front heat sink and the heat spreader between HF and LF boards. Meantime the assembly process of the HF/LF boards and the heat spreader in the middle enables a sufficient electromagnetic compatibility (EMC). The entire sensor has a compact size of $131.6\text{ mm} \times 107.5\text{ mm} \times 25.45\text{ mm}$, which is reasonably convenient for integration into various robotic platforms.

III. VERIFICATION RESULTS

A. Angular Resolution Test

To verify the angular separation capability of the implemented 4D radar sensor, two corner reflectors with radar cross section (RCS) value of 10 dBsm were placed each on the both sides along the boresight of the radar under test (RUT) in an anechoic chamber. The distance between RUT and two corner reflectors was set to 10 m with a calibrated optical bench. The minimum distance between two corner reflectors was noted, at which the RUT could still distinguish two corner reflectors as two separated groups of target points. The process was repeated for the test cases each on the azimuth and the elevation planes. Table 1 presents the tested angular separation results for the mentioned test cases. The designed angular separation of 1.5° on the azimuth plane and 3° on the elevation plane have been well realized.

B. Field-of-View (FoV)

To verify the FoV of the implemented 4D radar sensor, static targets with various RCS values of -5dBsm, 0 dBsm, 10 dBsm, and 20 dBsm were set for extracting maximum detection range at various incident angles during the measurement. The maximum detection range values were noted only when a stable detection behavior was confirmed (at least 90% detection statistics verified by the data flow). The verified FoV diagrams each of azimuth and elevation plane are presented in Fig. 4 with accordingly RCS values of -5dBsm, 0 dBsm, 10 dBsm, and 20 dBsm, respectively.

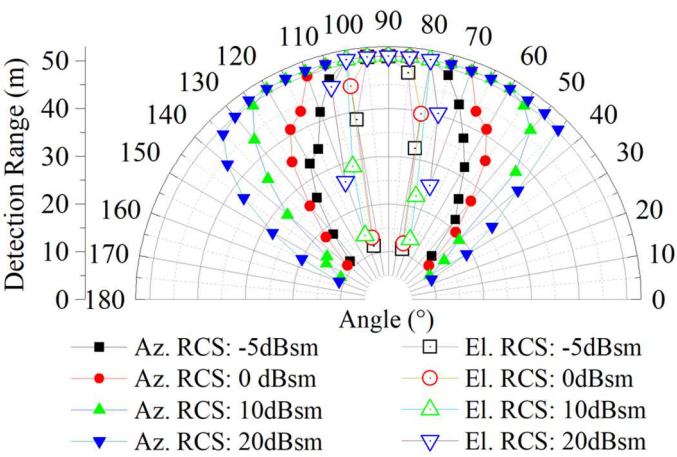


Fig. 4. Tested FoV of the implemented 4D mm-wave imaging radar sensor each on the azimuth (blue square point-line) and on the elevation (red triangle point-line) plane using a target with RCS values of -5 dBsm (black), 0 dBsm (red), 10 dBsm (green), and 20 dBsm (blue).

C. Indoor SLAM Demonstration

The feasibility of indoor radar-based SLAM was demonstrated in a proof-of-concept scenario using an indoor three-wheel hospitality robot as show in Fig. 5. The robot was remote controlled by a test engineer with a typical speed of 1-2 m/s. The indoor test location was selected to resemble the typical features of industrial workshops (metal shelves, industrial machines, pillars) and tradeshow halls (chairs, podiums, tall tables) to address potential future use cases.



Fig. 5. Demonstration scenario and setup of the indoor radar-based SLAM using a hospitality robot platform.

The robot was equipped with a stereo RGB camera (30 Hz), a solid-state scanning lidar (10 Hz), and three in this work implemented 4D radar sensor prototypes (10 Hz) mounted in a way that one points along the front of the ego vehicle, while the other two to the rear-left and -right around $\pm 120^\circ$ respectively. This surround radar view ensures that at least one of the sensors have visibility over (a part of) the scene even when the robot is close to walls or other large objects on one side. All sensors were mounted on a metallic pole structure (Fig. 5) to give them better visibility over the chassis of the robot. All sensors were calibrated, synchronized based on network time protocol (NTP), and integrated into the robotic operating system (ROS) for easier sensor management, visualization, and data recording.

Since global navigation satellite system (GNSS) signal is not available indoors as reference, lidar-based SLAM was chosen to create the reference trajectories for the proposed radar-based SLAM solution. To do this, the widely used Kiss-ICP SLAM method [18] was applied on the output from the lidar sensor mounted on the test robot. While the results are plausible, it is worth mentioning that the trajectories obtained with this method should only be considered as references instead of ground truth. A good illustration of this can be seen in Fig. 6, where the lidar map is imperfect in the top left corner. Possible reasons for this could be the limited FoV of the used lidar sensor leading to improper point cloud registration when the sensor is too close to the walls or corners.

Indoor use of radar sensors is often limited by strong multipath effects caused by close-by, highly reflective vertical structures and the fact that such sensors are usually configured (maximum range, range resolution) for automotive driving scenarios. Thus, before applying algorithms to estimate the ego-motion of the sensor, it is critical to "clean" the radar point cloud as much as possible. To do this, filters were applied based

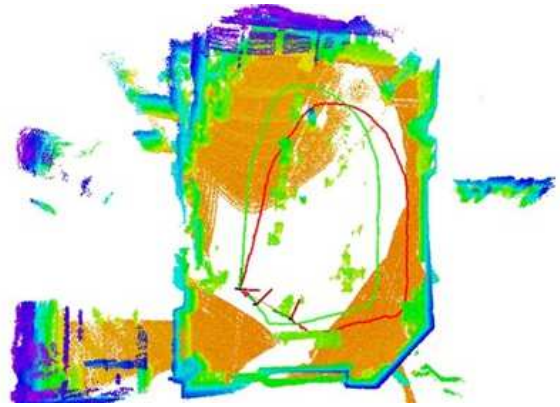


Fig. 6. Top view lidar map of the test scenario each with lidar- and radar-based trajectories marked with green (lidar-based) and red (radar-based) loops.

on distance (maximum 10 meters), reflectivity (minimum 35 signal-noise-ratio), and only points resulting from the peaks in the radar-cube are considered. Furthermore, all points were projected to a ground plane (i.e. assuming $Z = 0$) to simplify the regression of ego-motion. These parameters can be adjusted based on the target environment and the end use-case.

The scope of this work was to provide a proof-of-concept demonstration of the possibility of using 4D imaging radar sensors indoors for SLAM purpose. Thus, instead of developing novel methods, existing solutions were tried out on the pre-filtered radar point cloud. Given that the scenario only involved motion of the robot in planar environments without change in elevation, we assume zero translation in the vertical (Z axis) direction and zero roll and pitch for the ego-motion estimation. First, as a baseline, a basic point cloud matching using the Kabsch-Umeyama [19] algorithm between two consecutive scans is implemented. As seen in Fig. 7, the resulting trajectory (yellow) is far from the reference lidar (green) trajectory but indicates strong potential for point cloud matching approaches. Subsequently, the KISS-ICP [18] algorithm was modified to include the assumptions about ego-motion, i.e., only translation along longitudinal and lateral direction and yaw rotation is considered.

In Fig. 7, the red trajectory represents the output of this approach with much more accurate results than the baseline. Both methods are implemented on central processing units (CPUs) in real-time. As seen in Fig. 6 and Fig. 7, radar-based



Fig. 7. Qualitative comparison of the LiDAR-based reference trajectory (green), and the radar-based trajectories using basic point cloud matching with the Kabsch-Umeyama algorithm (yellow) and Kiss-ICP (red).

SLAM performs comparably to LiDAR-based SLAM for most parts of the trajectory. However, the performance drops in certain scenarios and motions. When the ego-vehicle is close to large artefacts and walls, the radar facing them cannot provide reliable point cloud output due to strong multipath and limited FoV. While the radars facing the other directions can help, the overall 360° point cloud becomes less salient for registration. This phenomenon is also observed while facing cornered walls where two out of the three sensors become momentarily quite noisy and thus the motion predicted during 90° turnings can be less accurate..

IV. CONCLUSION

A compact two-chip cascaded 4D imaging mm-wave radar sensor aiming for indoor 3D SLAM application is designed, implemented, and verified. By applying sparse antenna array and MIMO technique, 48 virtual channels are emulated, which can achieve angle separation each of 1.5° on azimuth and 3° on elevation plane. With maximum detection range of 51 m, the implemented 4D radar could well demonstrate a real-time mapping and localization effect in indoor scenarios. It is clear from the results that while adapting lidar-based solutions with minimal modifications gives promising results, there is room for improvement by addressing unique challenges and exploiting unique features offered by radars. First, in this work, the Doppler output of the radars was not utilized which could be a strong indicator for longitudinal translations and a signal to separate moving and static obstacles – a step critical in environments with humans. Second, it is well-known that the uncertainty in radars' angle estimation and radial velocity estimation depends on the angle-of-arrival and should be accounted for in the SLAM framework. Another consequence of these properties of radars is the poor rotation estimation, which could be addressed by introducing an inertial measurement unit (IMU). Finally, while the current setup with three radars provides a 360° view around the robot, it does not provide decent overlap between their field of views which could increase the robustness of selecting and matching stable points. In addition, the framework may also be adapted to ignore sensor(s) that are partially or fully blocked due to nearby walls, walking pedestrians, etc. leading to highly noisy point clouds. This allows for only stable and reliable radar detections to contribute to the localization and mapping efforts.

ACKNOWLEDGMENT

The authors wish to acknowledge other involved colleagues of R&D department of Nanjing Chuhang Technology Co., Ltd. (Team of Jie Zhang & Ji Zhang), P.R. China and entire manufacturing team of Anqing Chuhang Electronics Technology Co., Ltd. (Team of Xingfa Peng), P.R. China for their support.

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